



DIGITAL INDUSTRIES SOFTWARE

Thermo-mechanical stress on active chiplets in a 3D IC heterogeneous package assembly

Executive summary

The move to heterogeneous multi-chip/chiplet products improves yield, performance and modularity while reducing power and overall product footprint. However, this shift to heterogeneous assembly also introduces new complexities that can influence chip warpage and circuit behavior due to thermo-mechanical stress impacts.

In heterogeneous 3D IC architectures, the interaction between the chips and packages can compromise device functionality if not properly validated. Research has shown these thermo-mechanical stress impacts can lead to not only obvious package-level mechanical failures like die cracking, solder joint fatigue, warpage and delamination, but also more subtle impacts on circuit performance due to piezoresistive effects.

Evaluating and understanding the intra-package thermo-mechanical stress and its influence on circuit behavior is therefore crucial for ensuring the quality and reliability of heterogeneous 3D IC designs. This requires close collaboration between chip designers, package designers, and manufacturers to accurately model and simulate the 3D package assembly, including material properties and manufacturing processes.

By addressing stress-related issues early in the design cycle through a process of chip-package co-design and co-optimization, chip designers can mitigate the risks and ensure the performance and reliability of their heterogeneous 3D IC products.

Author: Shetha Nolke

Contents

Introduction: The rise of heterogenous 3D ICs

Risks of 3D IC stress

The impact of stress on 3D IC reliability

Evaluating stress in 3D IC packages

Interpreting 3D IC stress results

The path forward for 3D IC stress management

References

Introduction: The rise of heterogeneous 3D ICs

As we embark on the path to designing and manufacturing heterogeneous devices it will be important to eliminate systemic sources of failure in all three physical dimensions. It's well documented that this heterogeneous approach to new products will benefit from smaller IC design areas integrated by lower technology interposers and substrates. This dis-integrates the function from the technology in some sense, allowing the high function areas to be designed and manufactured on the leading-edge nodes while relying on established nodes for some of the integrative and less complex functions. This 3D IC innovation is going to have benefits for all markets from high performance compute to IoT devices.

Additionally, the same functioning can be packaged in smaller footprints and potentially reduce the power needs while increasing performance. Several benefits to moving to the heterogeneous design methodology have been proven and yet, there are more benefits yet to be discovered. Of course, building IC not only in 2D but adding on 2.5 to 3D elements will inherently introduce a host of challenges. Each product and design will have to make critical decisions as to where to trade off one benefit for another, or worse yet, one risk vs another risk. Some of the most common mechanical failures at 3D IC package level include die cracking, solder joint fatigue, warpage and delamination. Figure 1 illustrates die-level warpage. These types of failure are easy to detect but there are more insidious influences of stress at the device level that may only manifest as a failure to function as designed. The risk of electronic chip-package interaction causing device failure is a risk that chip designers need to be aware of, even early in the design cycle.

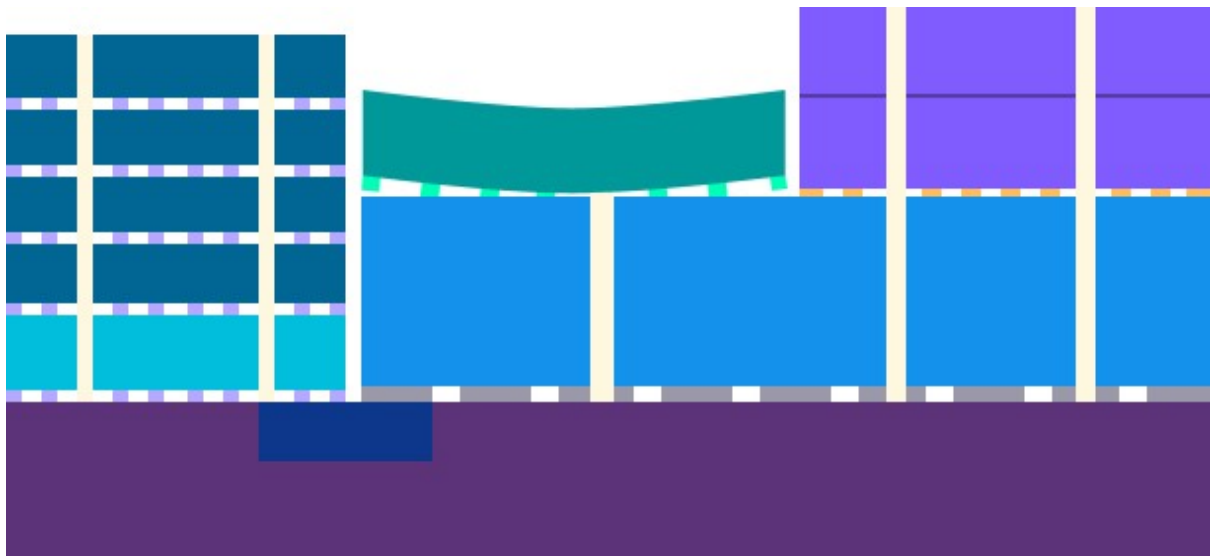


Figure 1. Die level warpage can cause lifting bumps and piezoresistive impacts.

Risks of 3D IC stress

Some risks have been identified early, forcing innovation in the way that these packages are designed and assembled. Most risks are still in discovery, as new configurations of assembly and package yield new ways for failures to occur. Probably the best understood concept of connecting devices and IC in 3D configuration is validation of design intent (DRC) and circuit intent (LVS) within and among various components. Less well defined is the understanding of how the power input may be converted into heat, and how heat may or may not be able to disperse within the devices and package. Another area of thermo-mechanical impact occurs during the chip packaging flow/reflow, where the temperature gaps exceed both typical SoC processing temperatures and power-induced thermal loads. These unavoidable thermal loads, thinned dies and the nature of the metal/dielectric material properties combine to exacerbate the risk of mechanical failure within the die. As with any failures, mechanical changes in the package don't only imply mechanical impacts but also imply electrical influences due to well defined piezoresistive effects [1].

The impact of stress on 3D IC reliability

While all circuits integrated at the die level could be impacted by chip-package stress, certain types of devices that require extra reliability will be most at risk for failures. Specifically, devices which use piezoelectric changes to determine device function will require early modeling and validation that neither packaging process nor operating temperature ranges (which can range widely for automotive purposes) will put the device out of spec for piezoelectric mobility changes. This level of reliability will also be required for sensors integrated in health monitors, smartphones and even IoT, transportation and infrastructure devices. To ensure reliability it will be key to verify, both in early design phase and final sign-off mode, that the circuit level performance will not be negatively impacted by chip/die level stress.

Evaluating stress in 3D IC packages

Evaluating 3D IC package stress seems intuitive in theory – how difficult could it be to calculate a simple physical model of stress? In fact – the physics component of this problem is the easiest part to solve. But before any physical computations can be made, there is some due diligence to attend to. First priority is to describe the package in terms of physical components that can be evaluated. Package components can vary in size from the mm size down to the nm feature sizes involved in the chip design layout layers. As a chip designer, it will be essential to collaborate with the package design team, both at foundries and OSATs, to understand both package components and the packaging process conditions. This is where language becomes important, such as the 3Dblox standard of describing the package assembly components.

Packages that incorporate integrated circuit designs can vary from a simple single-die stack (2.5D) to quite complex multi-die assemblies (3D). All packages will be in some way analyzed for impacts due to stress for reliability purposes. Some part of the package evaluation may need to be done before final decisions have been made about some package components. As such, some sort of modularity of information will need to occur to get as accurate of a description as possible to evaluate the stresses within the package. This means that early in the pathfinding stage of design, there may be assumptions about materials and component sizes that may change due to the learnings of these early-stage simulations. Even the die-level design may not be fully completed, but knowledge of the design technology (layers, thicknesses, density specs and routing direction) can be used to create an accurate initial estimate of stress profiles across the die. Later on, as package component designs are finalized, the package description can be updated with detailed material and size information for accurate stress modeling. This process is commonly referred to as chip-package codesign and may take months or even years, depending on the package complexity. Figure 2 illustrates the levels of 3D design from chiplet to SoC to board.

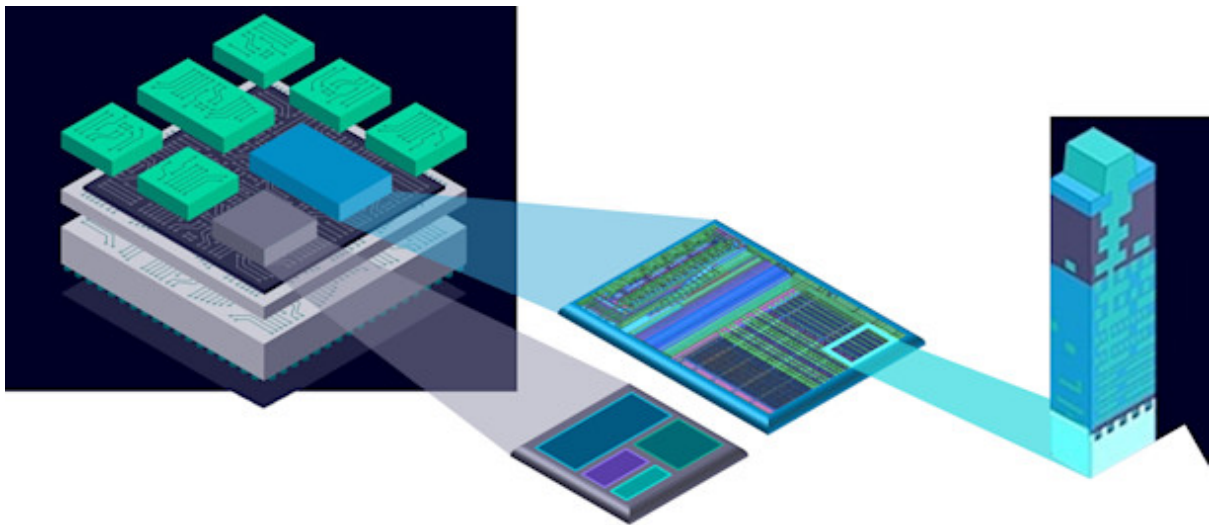


Figure 2. 3D IC assemblies need to be evaluated for stress at multiple levels of integration.

Another consideration would be who is evaluating the package stress? Is it the IC designer? The packaging designer? Each will have their own context from which to understand the risks for failure in different parts of the product. The solution space for stress evaluation should consider all these cases in a collaborative co-optimization.

The next consideration of evaluating stress is the physical material properties of the components within the assembly. With respect to the content of the package, any one person or group may only fully understand their portion of the package. Describing the material properties for any component can be as simple as assuming a single material “black box” for unknown components. This was a common practice for the IC, to assume a silicon block. This type of assumption will result in stress values, but the stress computations will be less accurate than describing the IC stack with more detailed physical description. Another point of concern might be that the technology description and material properties might be a foundry or packaging team’s intellectual property and require obfuscation.

Beyond the assembly definition and material properties, it’s important to fully understand how the packaging process itself imparts stresses within the package (figure 3). Especially within the IC portions of the package, adjacent materials can, and most likely will, behave very differently as temperatures change. Temperature changes in the package can occur at different stages of assembly and at different magnitudes. To determine more accurate stress values, one should consider nominal temperatures vs peak or trough values, such as annealing temperatures, or locally in area of soldering of BGA. Once a designer has described the package and properties to the best of their ability, finite element analysis can be used to evaluate the resulting stress within the package.

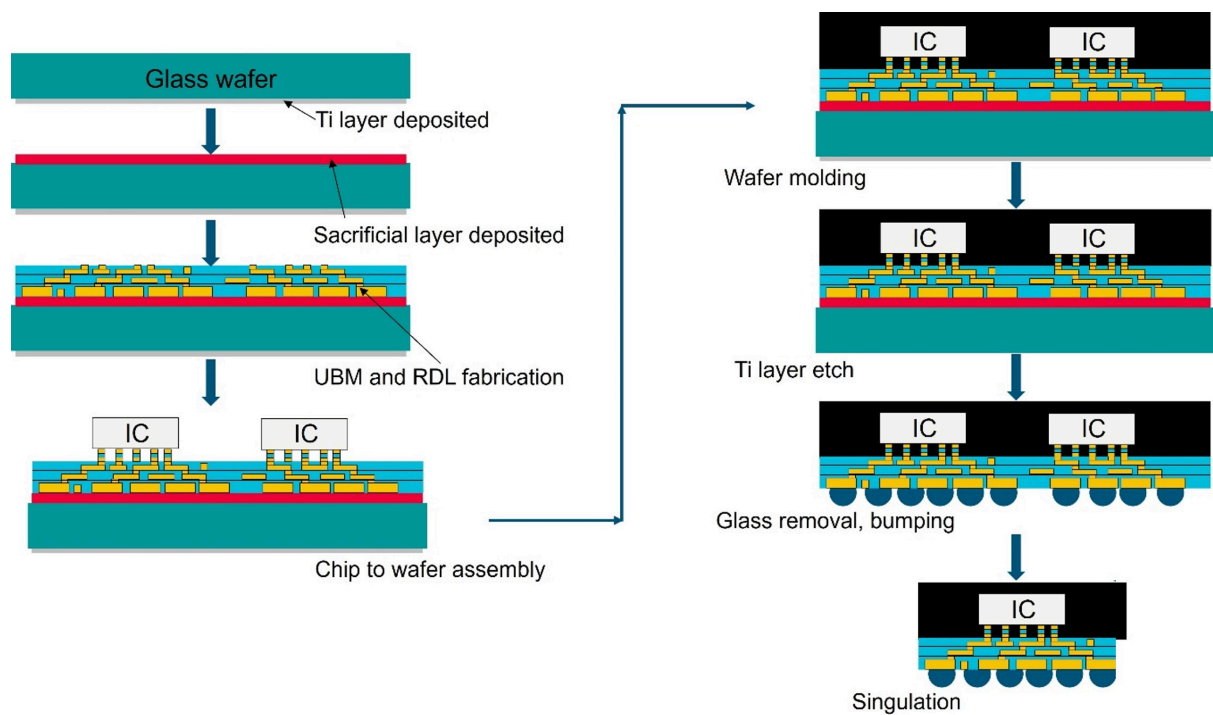


Figure 3. Stresses can be imparted throughout the 3D IC package assembly flow.

Interpreting 3D IC stress results

After initial boundaries are defined and the package stresses are evaluated, a designer has several choices to make with the resulting stress values. An obvious first step would be to quickly assess whether basic stress thresholds for relevant materials have been exceeded. This could be as simple as go/no-go or as complex as re-arrange and re-simulate iterative analysis. Early design full-package analysis can be done quickly and deliver informative results if reasonable assumptions are made. A chip level designer would be wise to carefully consider package composition early while several degrees of freedom still remain available.

Further along in the package planning and design, components will be locked in place and chiplets/IC can be considered in greater detail. At this stage, perhaps interfaces between package and chiplets can be evaluated for stresses which might indicate regions that critical devices should avoid. Given some detail about device location within the chiplet, a “what-if” analysis may be needed to indicate whether specific placements of the macros within the chiplet will exceed mechanical or electrical impact thresholds for these critical devices based on location in the package. After final locations of all components are decided, a sign-off analysis for chip-package interaction stresses will validate that circuit performance is within acceptable tolerances to ensure circuit reliability for the completed product. Figure 4 shows stress distribution map views.

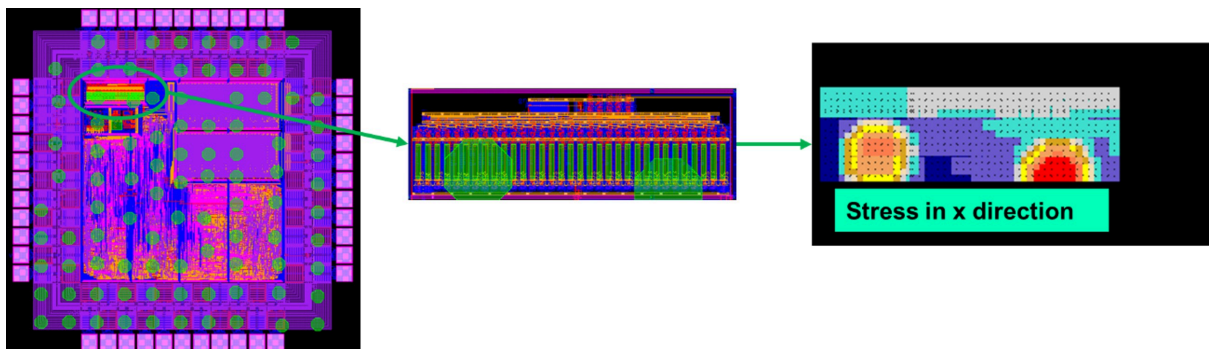


Figure 4. Stress distribution maps.

Aside from initial go/no go results dispositioning, other methods may need to be considered. Visual results based on specific regions can be evaluated for visual inspection of areas of concern. This can be done in very detailed areas or might be better suited to a global analysis, or perhaps somewhere in between at a package level interface. In addition to normal stress values, shear stress, warpage/offset and even mobility may be considered in a visual fashion.

These types of analysis all offer unique information about how stress effects will impact the region being simulated. Some focus more on overall quality of packaged IC and others speak to the reliability of the devices. Both are important for a successful product, in the end. In some cases, it has been shown that understanding

the piezoresistive behavior of stress on a device can be modeled with respect to circuit behavior and considered in the LVS extraction verification by way of back-annotation of stress values to the schematic netlist devices [2].

The path forward for 3D IC stress management

So, what's next? Clearly the shift to heterogeneous design and advanced packaging has introduced a level of complexity beyond technologies that we have seen before. This was also true of all new technologies since the introduction of the point contact transistor over a century ago. Physics has been in the driver's seat along the entire journey and it's not soon likely to give up the wheel. As we navigate this new terrain in 3D, the complexity of physical verification will require the ingenuity of many to converge on stable, reliable and producible results, for both inspection and manufacturing of 3D IC products. For both mechanical and electrical reasons, physical stress will become a major player in the game of successful design, both at die and package level.

References

1. Bridgman PW. Some Properties of Single Metal Crystals. Proc Natl Acad Sci U S A. 1924 Oct;10(10):411-5. doi: 10.1073/pnas.10.10.411. PMID: 16576849; PMCID: PMC1085739.
2. Kteyan A., et. al. Novel Methodology for Assessing Chip-Package Interaction Effects on Chip Performance. ISPD '22: Proceedings of the 2022 International Symposium on Physical Design:83-89. doi: <https://doi.org/10.1145/3505170.3511798>

Siemens Digital Industries Software

Offices

Headquarters

Granite Park One 5800 Granite
Parkway
Suite 600
Plano, TX 75024 USA
+1 972 987 3000

Americas

Granite Park One 5800 Granite
Parkway
Suite 600
Plano, TX 75024 USA
+1 314 264 8499

Europe

Stephenson House
Sir William Siemens Square Frimley,
Camberley
Surrey, GU16 8QD
+44 (0) 1276 413200

Asia-Pacific

Unit 901-902, 9/F
Tower B, Manulife Financial Centre
223-231 Wai Yip Street, Kwun Tong
Kowloon, Hong Kong
+852 2230 3333

For additional numbers, [click here](#).

About us

Siemens Digital Industries Software helps organizations of all sizes digitally transform using software, hardware and services from the Siemens Xcelerator business platform. Siemens' software and the comprehensive digital twin enable companies to optimize their design, engineering and manufacturing processes to turn today's ideas into the sustainable products of the future. From chips to entire systems, from product to process, across all industries. [Siemens Digital Industries Software](#) – Accelerating transformation.

Shetha Nolke